The dihedral head cannot meet a natural alternating tail

Human Compression project: bounded L044 proof component

All permutations act on the right; $g^h = h^{-1}gh$. A subgroup described as *pure* on a set fixes its complement pointwise. Products and intersections below are of actual subgroups, not of abstract isomorphism types or unproved projected C-markings.

Lemma 1 (A seven-point alternating subgroup)**.** *Let X be disjoint from $\{1,2,3,4,5\}$ and contain $6, y$, with $y \neq 6$. Put*

$$l = (1\ 2)(3\ 4), \qquad s = (2\ 5)(1\ 3), \qquad t = (1\ 3)(2\ 4)(5\ 6)\delta,$$

where δ fixes $1, \dots, 6$ and is supported on X. Let $e = (6\ y)u$ be an involution supported on X, where u fixes $6, y$. Then $\langle l, s, t, e\rangle$ contains the pure alternating group on $Z = \{1,2,3,4,5,6,y\}$.

Proof. The subgroup $D = \langle l, s, s^t\rangle$ is pure on $Z_0 = \{1, \dots, 6\}$. Indeed $s^t = (4\ 6)(1\ 3)$. The group $\langle l, s\rangle$ is transitive on $\{1, \dots, 5\}$: its product is a 5-cycle. Since s^t moves 6, D is transitive on Z_0. Its stabilizer of 6 contains $\langle l, s\rangle$ and is transitive on the other five points. Thus D is 2-transitive on Z_0.

Put $a = s^t$. One has $a^e = (4\ y)(1\ 3)$, so aa^e is a pure 3-cycle on $\{4, 6, y\}$. Conjugating it by D, which fixes y, gives the 3-cycles on $\{v, w, y\}$ for all ordered distinct $v, w \in Z_0$. These generate $\mathrm{Alt}(Z)$: they include every 3-cycle involving y, and products of two such cycles express those not involving y. All these operations occur inside $\langle l, s, t, e\rangle$. □

Theorem 2 (The remaining decorated dihedral case is impossible)**.** *Let $G = \langle \rho_0, \dots, \rho_{r-1}\rangle \leq \mathrm{Alt}(\Omega)$ be a string C-group, where $\Omega = \{1,2,3,4,5\} \sqcup X$ and $|X| \geq 5$. Suppose its actual first three generators are*

$$\rho_0 = l = (1\ 2)(3\ 4), \qquad \rho_1 = s = (2\ 5)(1\ 3), \qquad \rho_2 = t = \lambda(5\ 6)\delta, \quad \lambda = (1\ 3)(2\ 4),$$

where $6 \in X$, and δ is an involution supported on $X \setminus \{6\}$. Suppose $K = \langle \rho_3, \dots, \rho_{r-1}\rangle$ is the full pure natural alternating group $\mathrm{Alt}(X)$. No such marking exists.

Proof. Since λ is even and t is even, δ is odd. Every ρ_j with $j \geq 4$ fixes 5 and commutes with t, so fixes $6 = 5^t$ and centralizes δ. Transitivity of K therefore forces $e = \rho_3$ to move 6. Write $e = (6\ y)u$, where $y \in X \setminus \{6\}$ and u is an odd involution on $X \setminus \{6, y\}$.

We first identify an actual subgroup used in the intersection argument. Set $B = \{5\} \cup X$ and $R = \langle t, K\rangle$. Then $R \leq \langle\lambda\rangle \times \mathrm{Alt}(B)$. Its projection to B is the full $\mathrm{Alt}(B)$: it contains $\mathrm{Alt}(X)$ and its conjugate under $(5\ 6)\delta$, namely the alternating group on $B \setminus \{6\}$. These two natural alternating groups, on sets of size at least five with all but one point in common, generate $\mathrm{Alt}(B)$ by their 3-cycles. The other projection is onto $\langle\lambda\rangle$. If the projection to $\mathrm{Alt}(B)$ were injective, it would give a nontrivial homomorphism from the perfect group $\mathrm{Alt}(B)$ to C_2. This is impossible. Its kernel is consequently $\langle\lambda\rangle$, and

$$R = \langle\lambda\rangle \times \mathrm{Alt}(B).$$

Let $P = \langle l, s, t, e\rangle$ and $Z = \{1, \dots, 6, y\}$. By Lemma 1, P contains pure $\mathrm{Alt}(Z)$. In particular it contains both λ and the pure 3-cycle $c = (5\ 6\ y)$. Both elements also lie in R. Full ambient IP gives

$$P \cap R = \langle t, e\rangle.$$

This is a finite dihedral group, in which all elements of odd order belong to the cyclic rotation subgroup and commute. If δ moved y, the two 3-cycles c, c^t would have supports $\{5, 6, y\}$ and $\{5, 6, y^\delta\}$, respectively, with exactly two common points. They do not commute. Both belong to $\langle t, e\rangle$, a contradiction. Hence $y^\delta = y$.

It follows that P preserves Z. Any element in the kernel of its action on Z is supported on $X \setminus \{6, y\}$ and is even, since $G \leq \mathrm{Alt}(\Omega)$. It therefore belongs to $K = \mathrm{Alt}(X)$. But full ambient IP also gives

$$P \cap K = \langle e\rangle.$$

The nonidentity element e moves 6 and y, so this kernel is trivial. Thus the action of P on Z is faithful. It contains pure $\mathrm{Alt}(Z)$, and t induces the odd permutation $\lambda(5\ 6)$, so its image is $\mathrm{Sym}(Z)$.

In particular every element of P inducing an even permutation on Z is the unique pure alternating permutation with that action. Apply this to te: both t and e are odd on Z, so te acts trivially off Z. Their exterior parts are δ and u, respectively, whence $\delta u = 1$ and $u = \delta$. Consequently $e = (6\ y)\delta$ centralizes δ, which already centralizes every other generator of K. The permutation δ centralizes the transitive group K and fixes 6, so it fixes all of X. This contradicts its oddness and proves the theorem. □

Source application and limits. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. This fills the remaining natural-alternating-tail case in Proposition 5.14 (physical pages 28–29) and the $k = 2$ residual isolated in the accepted replacement of the intransitive-side equality step of Proposition 5.26 (physical pages 33–34). The two descriptions have the same literal marking after relabelling. The dihedral head, the source's attachment strategy and the full-IP requirement are inherited; the seven-point alternating-subgroup and faithful kernel argument above are the present replacement proof.

The six-point projection of $\langle l, s, t\rangle$ is the decorated S_5 case identified in the earlier local audit; it must not be silently replaced by either A_5 row of source Table 2. Our proof does not need that structural identification. It uses no smaller-degree rank induction, no projected IP, no finite census and no new formal theorem. It does not by itself settle the other source branches or confer whole L044 recovery or question credit.